

Effect of a $\geq 50\%$ Reduction in Low-Density Lipoprotein Cholesterol From Baseline in Patients With Ischemic Stroke

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Abstract

Background and Objectives

Current secondary atherosclerotic cardiovascular disease prevention guidelines recommend both a $\geq 50\%$ reduction in low-density lipoprotein cholesterol (LDL-C) from baseline and an absolute LDL-C goal (<70 or <55 mg/dL). Stroke-specific guidelines primarily emphasize only an absolute LDL-C goal, and the independent clinical impact of achieving a $\geq 50\%$ reduction remains unclear in patients with ischemic stroke. This study aimed to evaluate whether achieving a $\geq 50\%$ LDL-C reduction provides prognostic benefit independent of attaining an absolute LDL-C goal in patients with ischemic stroke.

Methods

This retrospective cohort study included patients admitted for acute ischemic stroke (2014–2022) using the Korean National Health Insurance Database in South Korea. LDL-C status over time was derived from serial national health examinations following the index stroke. Patients were categorized into 4 groups according to attainment of absolute (<70 mg/dL) and relative ($\geq 50\%$ reduction from baseline) LDL-C goals. The primary outcome was a composite of recurrent stroke, myocardial infarction, and all-cause death, analyzed using time-varying Cox regression.

Results

Among 89,414 patients, 136,427 poststroke LDL-C measurements were analyzed over a mean follow-up of 5.72 ± 2.40 years. Throughout the follow-up period, the proportion of patients achieving a $\geq 50\%$ LDL-C reduction from baseline remained at approximately 30%. Using the optimal group (achieving both <70 mg/dL and $\geq 50\%$ reduction) as the reference, the risk of the primary outcome was significantly higher in groups failing to achieve a $\geq 50\%$ reduction, regardless of attainment of the absolute LDL-C level (<70 mg/dL): adjusted hazard ratio [95% CI] 1.12 [1.06–1.18] for the <70 mg/dL and $<50\%$ reduction group and 1.28 [1.23–1.34] for the ≥ 70 mg/dL and $<50\%$ reduction group. Results were consistent across subgroups stratified by sex, age, baseline LDL-C level, and the presence of presumed cardioembolic source.

Discussion

Achieving a $\geq 50\%$ reduction in LDL-C from baseline was independently associated with a lower risk of secondary cardiovascular events. A substantial proportion of patients with ischemic stroke fail to achieve optimal LDL-C reduction; thus, more intensive and multifaceted lipid-lowering strategies focusing on both absolute and relative LDL-C reduction may lead to improved long-term outcomes.

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Supplementary Material

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Glossary

ASCVD = atherosclerotic cardiovascular disease; **AHA** = American Heart Association; **ASA** = American Stroke Association; **ESC** = European Society of Cardiology; **EAS** = European Atherosclerosis Society; **ESO** = European Stroke Organization; **HDL-C** = high-density lipoprotein cholesterol; **HR** = hazard ratio; **ICD-10** = International Statistical Classification of Diseases and Related Health Problems, 10th Revision; **LDL-C** = low-density lipoprotein cholesterol; **MET** = metabolic equivalent of task; **MI** = myocardial infarction; **NHIS** = National Health Insurance Service; **RR** = relative risk; **SPARCL** = Stroke Prevention by Aggressive Reduction of Cholesterol Levels; **TIA** = transient ischemic attack; **TST** = Treat Stroke to Target.

Introduction

Low-density lipoprotein cholesterol (LDL-C) is a well-established causal risk factor for atherosclerotic cardiovascular disease (ASCVD) including ischemic stroke.¹⁻³ Growing evidence from randomized clinical trials and epidemiologic studies has documented a dose-dependent, log-linear relationship between LDL-C levels and the risk of ASCVD events, reinforcing the concept “the lower, the better.”^{4,5} Accordingly, the 2026 American College of Cardiology/American Heart Association (ACC/AHA) and 2019 European Society of Cardiology/European Atherosclerosis Society (ESC/EAS) guidelines recommend intensive LDL-C-lowering therapy to achieve both a relative reduction of $\geq 50\%$ from baseline and an absolute LDL-C goal (< 70 or < 55 mg/dL) in patients with established ASCVD for secondary prevention.⁶⁻⁸

Although ischemic stroke represents a major clinical manifestation of ASCVD, evidence supporting the benefit of achieving a $\geq 50\%$ reduction in LDL-C from baseline remains relatively limited in this population. In the context of secondary prevention, current stroke-specific guidelines, including the 2021 AHA/American Stroke Association (ASA) and 2022 European Stroke Organization (ESO) guidelines, primarily emphasize intensive statin therapy aimed at achieving an LDL-C goal of < 70 mg/dL.^{9,10} Crucially, however, these guidelines do not provide specific recommendations regarding the magnitude of relative LDL-C reduction from baseline.

Given the robust evidence indicating that achieving a $\geq 50\%$ reduction in LDL-C from baseline is associated with a lower risk of future cardiovascular events in patients with coronary artery disease and other high-risk populations,¹¹⁻¹⁴ this study aimed to evaluate the independent prognostic impact of achieving a $\geq 50\%$ reduction in LDL-C beyond the attainment of an absolute LDL-C level of < 70 mg/dL in patients with ischemic stroke.

Methods

Standard Protocol Approvals, Registrations, and Patient Consents

This study was approved by the Institutional Review Board of Yongin Severance Hospital, Yonsei University Health System (no. 9-2024-0145). The requirement for informed consent was waived because of the retrospective nature of the study and the use of an anonymous health insurance claims database.

Study Design and Data Sources

This retrospective cohort study included patients with acute ischemic stroke using the nationwide health claims and health examination database in South Korea. South Korea operates a single-payer public health insurance system through the National Health Insurance Service (NHIS). The NHIS claims database contains comprehensive demographic information and medical resource utilization data, including hospital visits, diagnoses, medical procedures, and prescription records, for the entire Korean population. Diagnoses are recorded at each hospital visit according to the ICD-10. To promote public health, the NHIS provides a free national health examination program every 2 years. The participation rate in this program was approximately 70–80% between 2017 and 2019.¹⁵ The NHIS provides anonymized claims and health examination data for academic and policy research purposes.¹⁶ This database was used to evaluate the association between poststroke LDL-C levels relative to baseline and subsequent cardiovascular outcomes in the Korean population.

Study Participants

From the NHIS claims database, patients admitted with acute ischemic stroke between 2014 and 2022 were selected. Acute ischemic stroke was defined by admission of ≥ 3 days with a primary diagnosis code of I63, accompanied by brain computed tomography or magnetic resonance imaging during the hospitalization.^{17,18} The index date was defined as the date of admission for the qualifying ischemic stroke event.

To evaluate the effect of achieving a $\geq 50\%$ reduction in LDL-C from baseline, only patients who met the following criteria were included in the study:

1. Availability of a baseline LDL-C values obtained from the national health examination program during the pre-stroke period (from January 2009 to 1 month before the index date), with no use of statins or ezetimibe at the time of examination.
2. Availability of at least 1 poststroke LDL-C measurement recorded between 1 month after the index date and June 2023. Patients who experienced a primary outcome or who were lost to follow-up before the first poststroke health examination were excluded.

To ensure the inclusion of incident acute ischemic stroke cases, individuals with a history of stroke, defined by ICD-10

codes of I60–64, I69 in the health claims database or who self-reported stroke history in the health examination during the washout period from January 2009 to the index date were excluded. Patients with missing covariates or incomplete health examination data were also excluded. Figure 1 shows a flow chart of the selection of study participants.

Covariates Including LDL-C Levels

Sex and age at the index stroke were collected as time-invariant covariates. Since the national health examination is recommended biennially for the entire Korean adult population, longitudinal poststroke LDL-C measurements were obtained. Repeat measurements of poststroke LDL-C levels were treated as time-varying covariates and were categorized into 4 groups according to the achievement of absolute and relative goals: (1) $\geq 50\%$ reduction and < 70 mg/dL, (2) $\geq 50\%$ reduction and ≥ 70 mg/dL, (3) $< 50\%$ reduction and < 70 mg/dL, and (4) $< 50\%$ reduction and ≥ 70 mg/dL. The percentage reduction of LDL-C (range, 0%–100%) was calculated as follows:

$$\frac{\text{baseline LDL-C} - \text{post-stroke LDL-C}}{\text{baseline LDL-C}} \times 100$$

If poststroke LDL-C exceeded the baseline level, the percentage reduction was defined as 0%. Furthermore, in accordance with the 2026 ACC/AHA and 2019 ESC/EAS guideline recommendations for patients with clinical ASCVD at very high risk,^{6,7} we additionally categorized the poststroke LDL-C measurements into 4 analogous groups using a more stringent absolute LDL-C goal of < 55 mg/dL (instead of < 70 mg/dL) alongside the $\geq 50\%$ reduction target.

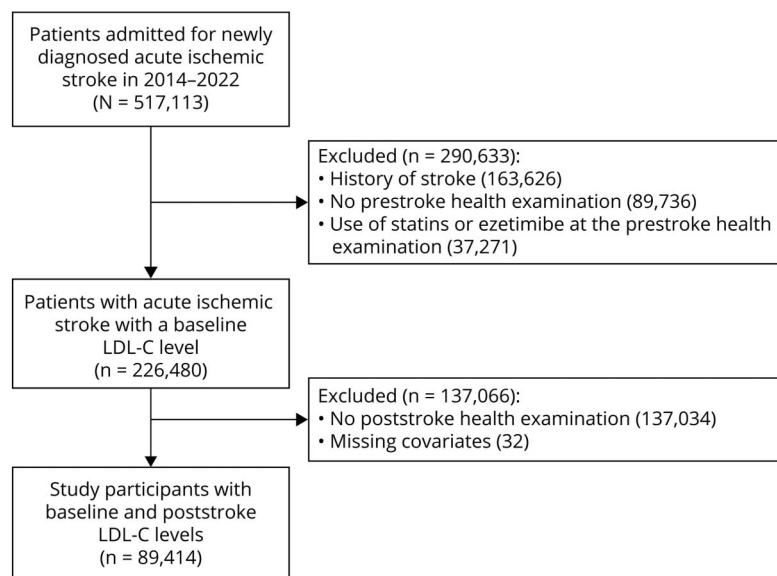
Along with LDL-C data, information on the degree of disability, household income (quartiles Q1–Q4), current

smoking, alcohol consumption, physical activity, comorbidities (hypertension, diabetes mellitus, atrial fibrillation, and chronic kidney disease), body mass index, and triglyceride levels was obtained from serial health examinations. These variables were treated as time-varying covariates, with values updated at each examination. The degree of disability was classified as normal, mild, or severe using data from the Korea National Disability Registration System, which determines disability status based on activities of daily living while considering individual socioeconomic factors.¹⁹ Detailed definitions of covariates are provided in eTable 1. As the primary objective of this study was to evaluate the effect of achieved LDL-C levels, the use of lipid-lowering medications during follow-up was not included as an adjustment variable.

Primary and Secondary Outcomes

The primary outcome was defined as a composite of recurrent stroke, myocardial infarction (MI), or all-cause death during follow-up. To avoid misclassifying chronic stroke or planned admissions for diagnostic evaluation as recurrent events, a recurrent stroke was defined as admission for ≥ 3 days or death with a primary diagnosis code of I60–I63 (I63 for ischemic stroke and I60–I62 for hemorrhagic stroke), accompanied by brain imaging.^{17,18} MI was identified by admission with primary diagnostic codes of I21–I22 or claims codes of percutaneous coronary intervention for MI (procedure codes M6553 or M6565). The diagnostic accuracy of stroke and MI based on NHIS claims data has been previously validated.^{17,20,21} Mortality data were obtained by linkage with Statistics Korea. Secondary outcomes included each individual component of the primary composite outcome (recurrent stroke, MI, and all-cause death). In addition, ischemic stroke alone and a composite outcome of ischemic stroke and MI were also evaluated as secondary outcomes.

Figure 1 Study Flowchart Showing Inclusion and Exclusion Criteria



LDL-C = low-density lipoprotein cholesterol.

Subgroup Analysis

To evaluate the potential effect modification, subgroup analyses were performed by stratifying patients by sex, age, and baseline LDL-C levels (≥ 140 mg/dL, 100 to <140 mg/dL, and <100 mg/dL). Because intensive lowering of LDL-C is recommended primarily for patients with ischemic stroke of atherosclerotic origin,¹⁰ patients with cardioembolic stroke may fall outside the intended scope of this strategy. To evaluate whether the association between LDL-C reduction and clinical outcomes differed according to presumed cardioembolic source, we also performed subgroup analysis stratified by the presence of presumed cardioembolic sources. Patients with potential cardioembolic sources were defined as those with atrial fibrillation (I48), rheumatic mitral valve disease (I05), sick sinus syndrome (I49.5), mechanical heart valve (Z95.2–Z95.4),²² or those receiving oral anticoagulation therapy (warfarin or direct oral anticoagulants) at the time of the first poststroke health examination.

Statistical Methods

Categorical variables are presented as numbers (%) and continuous variables as mean $\pm$ standard deviations. To evaluate the association between LDL-C categories and the risk of the primary outcome, we calculated adjusted hazard ratio (HR) and 95% CI using a multivariable time-varying Cox regression model. For the primary analysis, patients were categorized based on the absolute LDL-C target of <70 mg/dL. Furthermore, a supplementary analysis was conducted using a more stringent target of <55 mg/dL to align with contemporary guidelines for very-high-risk patients. The model was adjusted for sex, age, degree of disability, household income, current smoking, alcohol consumption, physical activity, hypertension, diabetes mellitus, atrial fibrillation, chronic kidney disease, body mass index, and triglyceride levels.

The proportional hazards assumption for the LDL-C categories in the multivariable Cox regression model was tested by calculating Schoenfeld residuals and the assumption was satisfied. To further explore the relationship between the percentage reduction in LDL-C from baseline and the risk of the primary outcome, a smooth HR curve was estimated using penalized splines implemented in the *smoothHR* package in R.²³ In addition, we examined the dose-response relationship according to quintiles and deciles of percentage LDL-C reduction. Quintiles and decile cutoffs were defined based on the percentage reduction from baseline to the first poststroke LDL-C measurement in the study population. Each poststroke LDL-C measurement was then categorized into the corresponding quintile or decile according to its percentage reduction from baseline using these predefined cutoffs. These categories were included in the analysis as time-varying variables, with the lowest reduction group, Q1 or D1, serving as the reference in each analysis.

Separate cause-specific Cox regression models were constructed to estimate adjusted HRs (95% CI) for individual secondary outcomes, including recurrent stroke, MI, a composite of ischemic stroke and MI, and all-cause death. All

analyses were performed using SAS v.9.4.2 (SAS Institute) and R v.4.0.3 (R Foundation for Statistical Computing). A 2-sided p value <0.05 was considered statistically significant.

Data Availability

The data set used in this study was provided by the NHIS of South Korea (nhiss.nhis.or.kr/bd/ab/bdaba000eng.do) specifically for this study (NHIS-2025-04-1-126) and is therefore not publicly available. Researchers may request access to data by submitting a research proposal, a completed application form, and valid institutional review board approval documents to the National Health Insurance Sharing Service (nhiss.nhis.or.kr) for review and approval.

Results

Patient Characteristics

From the NHIS claims database, we identified 517,113 patients admitted for acute ischemic stroke between 2014 and 2022. According to the inclusion and exclusion criteria, the study finally included 89,414 patients with acute ischemic stroke who had available baseline and poststroke LDL-C measurements (Figure 1). The characteristics of the study participants are summarized in Table 1. In this cohort, the mean age at the index stroke was 63.88 ± 11.38 years and the mean duration of follow-up after the index stroke was 5.72 ± 2.40 years.

Trends in Prestroke and Poststroke LDL-C Levels

During the follow-up period, 89,414 participants underwent a total of 136,427 poststroke health examinations with LDL-C measurements (1.53 ± 0.85 measurements per patient). The mean interval from the index stroke to the first poststroke LDL-C measurement was 1.85 ± 1.43 years. Figure 2A shows the distribution of baseline (measured before the index stroke) and poststroke LDL-C levels (mg/dL) throughout the follow-up period. The mean baseline LDL-C level was 123.56 ± 34.84 mg/dL. In the longitudinal analysis of poststroke LDL-C levels, the proportion of patients who achieved an LDL-C level of <70 mg/dL remained consistently $<50\%$ across the follow-up years.

Figure 2B illustrates the distribution of the relative reduction in LDL-C from baseline levels after the index stroke (% reduction). The proportion of patients achieving a $\geq 50\%$ relative LDL-C reduction was 34% within the first year after the index stroke and subsequently remained stable at approximately 30% throughout the poststroke period. When assessing the attainment of both goals, relative reduction ($\geq 50\%$ from baseline) and absolute level (<70 mg/dL) of LDL-C, the proportion of patients meeting both criteria remained at approximately 25% throughout the study period (Figure 2C).

Association Between $\geq 50\%$ Reduction in LDL-C and Clinical Outcomes

During the follow-up period, the primary outcomes occurred in 12,586 patients, including 5,645 cases of recurrent stroke,

Table 1 Clinical Characteristics of Study Participants

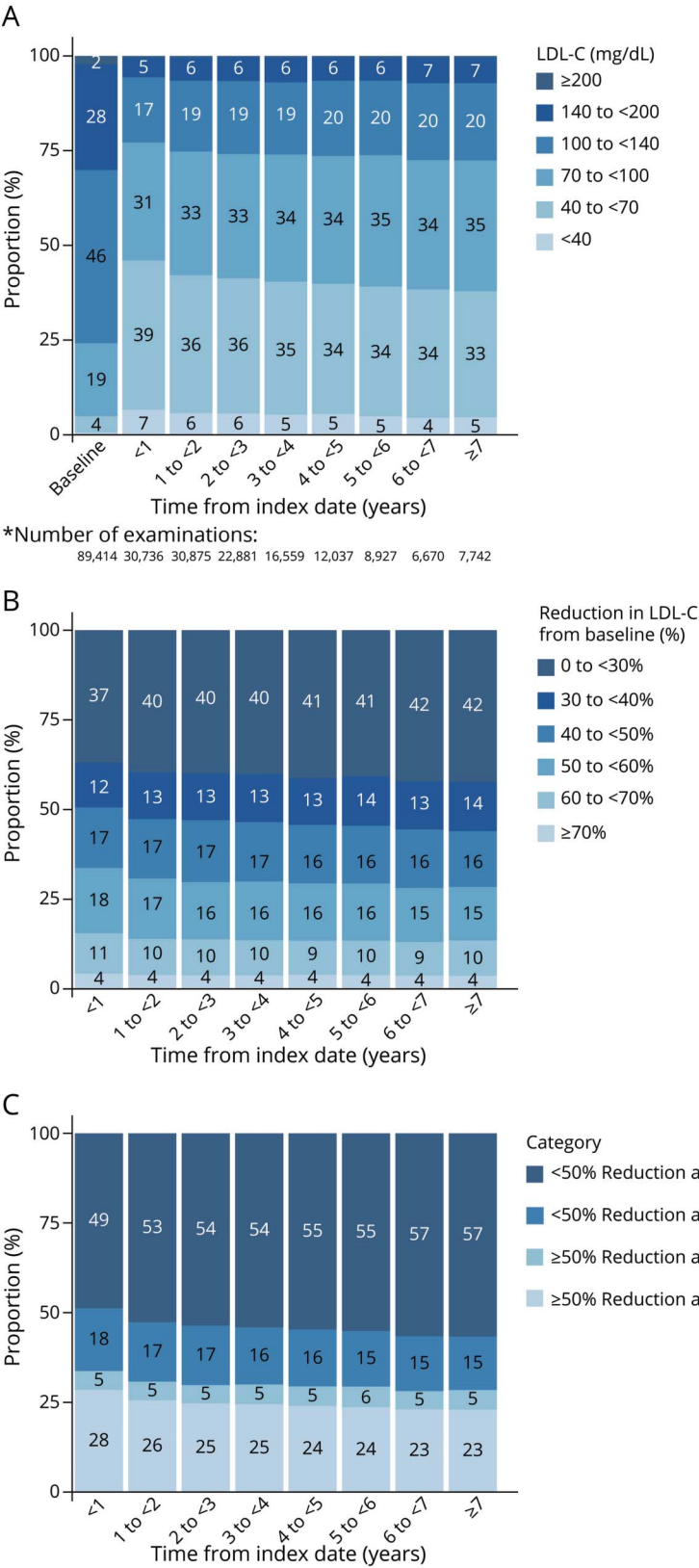
Variable	Value
Sex, male	58,960 (65.94)
Age, y	63.88 ± 11.38
Degree of disability	
Normal	75,387 (84.31)
Mild disability	3,050 (3.41)
Severe disability	10,977 (12.28)
Household income	
Q1, lowest	22,277 (24.91)
Q2	20,966 (23.45)
Q3	25,559 (28.59)
Q4, highest	20,612 (23.05)
Current smoking	14,523 (16.24)
Alcohol consumption, g/wk	
0	62,802 (70.24)
>0 to <105	18,691 (20.90)
105 to <210	4,783 (5.35)
≥210	3,138 (3.51)
Physical activity, MET-minutes/week	
0	32,303 (36.13)
>0 to <500	14,437 (16.15)
500 to <1,000	16,069 (17.97)
≥1,000	26,605 (29.75)
Comorbidities	
Hypertension	61,086 (68.32)
Diabetes mellitus	27,415 (30.66)
Atrial fibrillation	13,635 (15.25)
Chronic kidney disease	14,639 (16.37)
Anthropometric measurements and laboratory findings	
Body mass index, kg/m ²	24.52 ± 3.31
Triglyceride, mg/dL	121.24 ± 76.62
LDL-C, mg/dL	80.43 ± 33.41

Abbreviations: LDL-C = low-density lipoprotein cholesterol; MET = metabolic equivalent of task; Q = quartile. Values are presented as number (%) for categorical variables and mean ± SD for continuous variables. Age was assessed at the time of index ischemic stroke, and all other variables were obtained from the first national health examination conducted after the index ischemic stroke.

635 cases of MI, and 6,306 cases of all-cause death. We performed a time-varying Cox regression analysis for the primary outcome, categorizing patients into 4 groups based on their poststroke LDL-C status (Table 2). Compared with the reference group achieving both goals (≥50% reduction and <70 mg/dL), the group achieving only the relative

reduction (≥50% reduction and ≥70 mg/dL) showed no significant difference in risk (adjusted HR 1.02, 95% CI 0.92–1.12). By contrast, both groups that failed to achieve a ≥50% reduction, <50% reduction and <70 mg/dL (adjusted HR 1.12, 95% CI 1.06–1.18) and <50% reduction and ≥70 mg/dL (adjusted HR 1.28, 95% CI 1.23–1.34) had

Figure 2 Distribution of LDL-C Level Across the Study Period



Bars at each period represent the distribution of categories exclusively among the patients who underwent an LDL-C measurement during the corresponding period. (A) Baseline (prestroke) and poststroke LDL-C level (mg/dL), (B) magnitude of % reduction in LDL-C from baseline, and (C) accounting for both absolute level (LDL-C < 70 mg/dL vs ≥70 mg/dL) and relative reduction (≥50% reduction vs <50% reduction). *Number of LDL-C measurements during each period. LDL-C = low-density lipoprotein cholesterol.

a significantly increased risk of the primary outcome. Notably, the significant risk elevation in the group with <50% reduction and <70 mg/dL relative to the reference group suggests that achieving a $\geq 50\%$ reduction is independently important, even among ischemic stroke patients who successfully reach the absolute LDL-C level of <70 mg/dL. Regarding the secondary outcomes (Table 2), the risks for recurrent stroke, ischemic stroke, MI, ischemic stroke + MI, and all-cause death similarly increased in both groups failing to achieve a $\geq 50\%$ reduction in LDL-C level.

Furthermore, in alignment with the 2026 ACC/AHA and 2019 ESC/EAS guideline recommendations for patients with very-high-risk ASCVD,^{6,7} we performed an additional analysis using a more stringent absolute LDL-C target of <55 mg/dL (Table 3). In this analysis, the findings were highly consistent with those of the main analysis. Patients who achieved a $\geq 50\%$ reduction but did not reach the absolute goal ($\geq 50\%$ reduction and ≥ 55 mg/dL) showed no significant difference in risk (adjusted HR 1.01, 95% CI 0.94–1.08) compared with the reference group ($\geq 50\%$ reduction and <55 mg/dL). By contrast, both groups that failed to achieve a $\geq 50\%$ reduction demonstrated a significantly increased risk of the primary outcome: <50% reduction and <55 mg/dL (adjusted HR 1.13, 95% CI 1.03–1.23) and <50% reduction and ≥ 55 mg/dL (adjusted HR 1.25, 95% CI 1.19–1.32). These findings further suggest that failure to achieve a $\geq 50\%$ reduction in LDL-C is associated with increased risk, irrespective of attaining the absolute LDL-C goal.

When the percentage reduction in LDL-C was analyzed as a continuous variable in the model, each 10% reduction in

LDL-C from baseline was associated with an approximately 6% lower risk of the primary outcome (adjusted HR per 10% reduction 0.94, 95% CI 0.94–0.95). Consistently, the smooth spline curve demonstrated an inverse relationship, where the risk for the primary outcome decreased in proportion to the magnitude of the % reduction in LDL-C level (Figure 3). To further characterize this dose-response relationship, we categorized the percentage LDL-C reduction by quintiles and deciles. In both the quintile and decile analyses, greater LDL-C reduction was consistently associated with lower risk of the primary outcome, with the risk reduction plateauing at higher reduction categories, corroborating the pattern observed in the spline analysis (eFigure 1).

Subgroup Analyses

We performed subgroup analyses stratified by sex, age, and baseline LDL-C level (≥ 140 mg/dL, 100 to <140 mg/dL, <100 mg/dL). Across these subgroups, achieving a $\geq 50\%$ reduction in LDL-C was consistently associated with a lower risk of the primary outcome, with no significant interaction detected among subgroups (all p for interaction >0.1, Table 4). In the subgroup analysis excluding patients with presumed cardioembolic sources, the beneficial association of achieving a $\geq 50\%$ reduction in LDL-C persisted without attenuation (Table 4). In the complementary analysis restricted to patients with a presumed cardioembolic source, the association remained directionally consistent, although point estimates were modestly attenuated compared with the noncardioembolic subgroup (Table 4). No significant interaction was detected between presumed cardioembolic status and LDL-C reduction regarding the primary outcome (p for interaction = 0.298).

Table 2 Effect of Achieving $\geq 50\%$ Reduction in LDL-C and LDL-C <70 mg/dL on Clinical Outcomes in Patients With Ischemic Stroke

Outcomes	No. of patients with the outcome	Poststroke LDL-C			
		$\geq 50\%$ reduction and <70 mg/dL	$\geq 50\%$ reduction and ≥ 70 mg/dL	<50% reduction and <70 mg/dL	<50% reduction and ≥ 70 mg/dL
Primary outcome^a	12,586	1.00 (Ref)	1.02 [0.92–1.12]	1.12 [1.06–1.18]	1.28 [1.23–1.34]
Secondary outcomes					
Recurrent stroke	5,645	1.00 (Ref)	1.01 [0.88–1.16]	1.14 [1.04–1.24]	1.37 [1.28–1.47]
Ischemic stroke	4,949	1.00 (Ref)	1.01 [0.87–1.18]	1.11 [1.01–1.22]	1.41 [1.31–1.52]
MI	635	1.00 (Ref)	1.81 [1.25–2.63]	1.06 [0.79–1.41]	1.84 [1.49–2.27]
Ischemic stroke + MI	5,584	1.00 (Ref)	1.09 [0.95–1.25]	1.11 [1.01–1.21]	1.46 [1.36–1.56]
All-cause death	6,306	1.00 (Ref)	0.97 [0.84–1.11]	1.09 [1.01–1.18]	1.16 [1.09–1.23]

Abbreviations: HR = hazard ratio; LDL-C = low-density lipoprotein cholesterol; MI = myocardial infarction.

Data are presented as adjusted HR [95% CI] derived from multivariable time-varying Cox regression models. Models were adjusted for sex, age, degree of disability, household income, current smoking, alcohol consumption, physical activity, hypertension, diabetes mellitus, atrial fibrillation, chronic kidney disease, body mass index, and triglyceride levels.

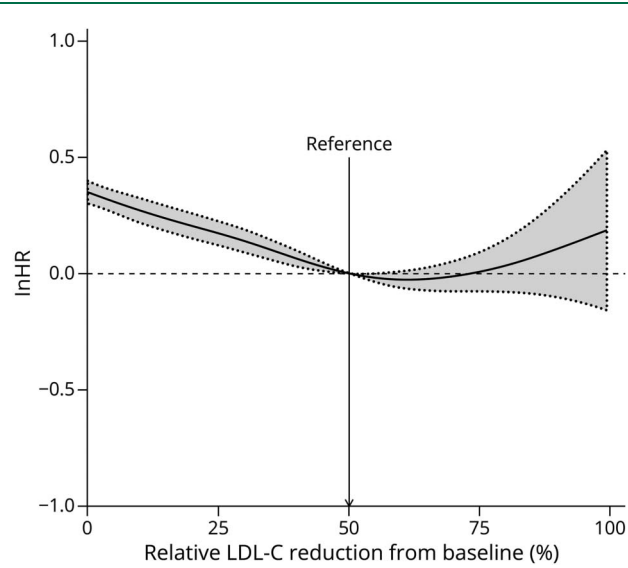
^aComposite development of recurrent stroke, MI, and all-cause death, whichever occurred earlier.

Table 3 Effect of Achieving $\geq 50\%$ Reduction in LDL-C and LDL-C < 55 mg/dL on Clinical Outcomes in Patients With Ischemic Stroke

Outcomes	No. of patients with the outcome	Poststroke LDL-C			
		$\geq 50\%$ reduction and < 55 mg/dL	$\geq 50\%$ reduction and ≥ 55 mg/dL	$< 50\%$ reduction and < 55 mg/dL	$< 50\%$ reduction and ≥ 55 mg/dL
Primary outcome ^a	12,586	1.00 (Ref)	1.01 [0.94–1.08]	1.13 [1.03–1.23]	1.25 [1.19–1.32]
Secondary outcomes					
Recurrent stroke	5,645	1.00 (Ref)	0.99 [0.90–1.10]	1.07 [0.93–1.23]	1.33 [1.23–1.44]
Ischemic stroke	4,949	1.00 (Ref)	1.01 [0.90–1.13]	1.08 [0.93–1.26]	1.36 [1.25–1.48]
MI	635	1.00 (Ref)	1.41 [1.02–1.94]	1.06 [0.66–1.68]	1.78 [1.39–2.29]
Ischemic stroke + MI	5,584	1.00 (Ref)	1.04 [0.94–1.16]	1.08 [0.94–1.25]	1.40 [1.29–1.52]
All-cause death	6,306	1.00 (Ref)	0.99 [0.90–1.09]	1.17 [1.04–1.33]	1.14 [1.06–1.23]

Abbreviations: HR = hazard ratio; LDL-C = low-density lipoprotein cholesterol; MI = myocardial infarction. Data are presented as adjusted HR [95% CI] derived from multivariable time-varying Cox regression models. Models were adjusted for sex, age, degree of disability, household income, current smoking, alcohol consumption, physical activity, hypertension, diabetes mellitus, atrial fibrillation, chronic kidney disease, body mass index, and triglyceride levels. ^aComposite development of recurrent stroke, MI, and all-cause death, whichever occurred earlier.

Figure 3 Smooth Hazard Ratio Curve Illustrating the Non-Linear Association Between Relative LDL-C Reduction From Baseline and the Risk of the Primary Outcome



The solid black line represents the estimated natural log hazard ratio (lnHR), and the gray shaded area indicates the 95% confidence intervals. The reference point was set at a 50% reduction in LDL-C from baseline. The curve was estimated using penalized splines within a multivariable Cox regression model for the primary outcome (*smoothHR* package in R). The model was adjusted for sex, age, degree of disability, household income, current smoking, alcohol consumption, physical activity, hypertension, diabetes mellitus, atrial fibrillation, chronic kidney disease, body mass index, and triglyceride levels. The y-axis indicates lnHR for the primary outcome, and the x-axis indicates the magnitude of %reduction in LDL-C from baseline. lnHR = natural log hazard ratio; LDL-C = low-density lipoprotein cholesterol.

Discussion

This cohort study evaluated the prognostic impact of a $\geq 50\%$ reduction in LDL-C from baseline on future cardiovascular outcomes in patients with ischemic stroke. Our analysis of serial national health examination data revealed that a substantial portion of patients maintained suboptimal LDL-C control after ischemic stroke. Specifically, only approximately 25% of patients achieved both the relative goal of a $\geq 50\%$ reduction and absolute goal of < 70 mg/dL throughout the follow-up period. It is important that achieving a $\geq 50\%$ reduction in LDL-C from baseline was independently associated with a reduced risk of recurrent cardiovascular events. These findings suggest that inadequate LDL-C control remains prevalent in clinical practice and support the potential value of considering achievement of a $\geq 50\%$ reduction in LDL-C from baseline as a key metric, beyond that of an absolute goal, in the optimal preventive strategy for patients with ischemic stroke.

LDL-C is a well-established causal risk factor for ASCVD. Numerous clinical trials and epidemiologic studies have clearly documented that LDL-C-lowering treatment effectively reduces the risk of cardiovascular events, and the risk reduction is more prominent with intensive LDL-C lowering compared with placebo or less intensive lowering.^{24–26} Trials using intravascular ultrasound in patients with coronary artery disease have demonstrated that an intensive LDL-C reduction of at least 50% from baseline with high-intensity statin therapy was significantly associated with a regression of coronary

Table 4 Subgroup Analyses of the Association Between Poststroke LDL-C Reduction and the Primary Outcome

Subgroups	No. of patients in the subgroup	Poststroke LDL-C			
		≥50% reduction and <70 mg/dL	≥50% reduction and ≥70 mg/dL	<50% reduction and <70 mg/dL	<50% reduction and ≥70 mg/dL
Baseline LDL-C level					
≥140 mg/dL	26,925	1.00 (Ref)	1.06 [0.95–1.18]	NA ^b	1.29 [1.19–1.40]
100 to <140 mg/dL	40,877	1.00 (Ref)	NA ^b	1.06 [0.97–1.15]	1.27 [1.19–1.35]
<100 mg/dL	21,612	1.00 (Ref)	NA ^b	1.05 [0.94–1.16]	1.24 [1.12–1.38]
Sex					
Male	58,960	1.00 (Ref)	1.03 [0.92–1.17]	1.14 [1.07–1.22]	1.33 [1.26–1.40]
Female	30,454	1.00 (Ref)	0.96 [0.83–1.12]	1.07 [0.96–1.20]	1.20 [1.11–1.30]
Age ^a					
<64	42,915	1.00 (Ref)	1.00 [0.84–1.19]	1.15 [1.03–1.30]	1.28 [1.17–1.40]
≥64	46,499	1.00 (Ref)	0.98 [0.87–1.10]	1.13 [1.06–1.20]	1.30 [1.23–1.37]
Presumed cardioembolic source					
Patients without a cardioembolic source	74,940	1.00 (Ref)	1.02 [0.92–1.13]	1.14 [1.07–1.21]	1.32 [1.25–1.38]
Patients with a cardioembolic source	14,474	1.00 (Ref)	1.02 [0.82–1.26]	1.05 [0.94–1.17]	1.18 [1.07–1.29]

Abbreviations: HR = hazard ratio; LDL-C = low-density lipoprotein cholesterol; NA = not applicable.
Data are presented as adjusted HR [95% CI] derived from multivariable time-varying Cox regression models for the primary outcome. Models were adjusted for sex, age, degree of disability, household income, current smoking, alcohol consumption, physical activity, hypertension, diabetes mellitus, atrial fibrillation, chronic kidney disease, body mass index, and triglyceride levels.
^aAge was dichotomized at the mean age at the index ischemic stroke.
^bNA indicates categories that were not applicable because such combinations could not occur given the baseline LDL-C levels.

atherosclerosis.^{27–29} In a pooled patient-level meta-analysis of 3 randomized trials with ASCVD with statin treatment, patients with a ≥50% reduction in LDL-C from baseline were at a reduced risk of cardiovascular events compared with those with <50% reduction in LDL-C regardless of attaining an LDL-C level of <70 mg/dL.¹² Based on the fundamental evidence of “lower is better,” intensive LDL-C-lowering treatment, aiming for at least a 50% reduction from baseline, is a cornerstone of cardiovascular preventive strategy for those with established ASCVD or at high-risk population.³⁰ Consequently, both the 2026 ACC/AHA and 2019 ESC/EAS guidelines universally recommended achieving a ≥50% reduction in LDL-C from baseline for patients with a history of ASCVD.^{6–8} Therefore, for patients with established ASCVD who fail to achieve a ≥50% reduction in LDL-C despite maximally tolerated statin therapy, additional lipid-lowering strategies are warranted.³¹ However, in contrast to these guidelines, the management of LDL-C often prioritizes meeting the specific target levels over achieving a ≥50% reduction from baseline in real-world clinical practice.³²

Regarding absolute LDL-C levels, the 2018 AHA/ACC guidelines recommend an LDL-C threshold of 70 mg/dL to guide therapy intensification.³³ The recently updated 2026 ACC/AHA guideline recommends a more stringent

absolute LDL-C goal of <55 mg/dL for secondary prevention in individuals with clinical ASCVD at very high risk.⁶ Similarly, the 2019 ESC/EAS guidelines recommend LDL-C targets of <55 mg/dL for patients with ASCVD, and even <40 mg/dL for those with recurrent ASCVD events.^{7,8}

For patients with ischemic stroke, intensive LDL-C-lowering treatment is also strongly recommended as secondary prevention,^{9,10} largely based on evidence from 2 pivotal randomized clinical trials.^{34,35} The Stroke Prevention by Aggressive Reduction of Cholesterol Levels (SPARCL) trial demonstrated that high-intensity statin therapy with atorvastatin 80 mg reduced the risk of major cardiovascular events by approximately 20% (adjusted HR 0.80, 95% CI 0.69–0.92) compared with placebo.³⁴ The Treat Stroke to Target (TST) trial showed that LDL-C-lowering treatment with a lower target (<70 mg/dL) resulted in a lower risk of subsequent cardiovascular events compared with a higher target range (90–110 mg/dL) in patients with a history of recent ischemic stroke or transient ischemic attack (TIA) and documented atherosclerosis.³⁵ In addition, a meta-analysis of 11 randomized clinical trials involving 20,163 stroke patients found that more intensive LDL-C-lowering statin-based therapies were associated with a reduced risk of recurrent stroke compared with less intensive therapies (absolute risk, 8.1% vs 9.3%;

relative risk (RR) 0.88, 95% CI 0.80–0.96).³⁶ Accordingly, the 2021 AHA/ASA guidelines for secondary stroke prevention recommend lipid-lowering therapy with a goal LDL-C of <70 mg/dL for patients with ischemic stroke or TIA and atherosclerotic disease,¹⁰ while also specifically recommending atorvastatin 80 mg daily in patients with LDL-C level >100 mg/dL. The 2022 ESO guidelines on pharmacologic interventions for long-term secondary prevention after ischemic stroke or TIA recommended an LDL-C target of <70 mg/dL,⁹ primarily citing the TST trial as the only head-to-head randomized comparison of LDL-C targets.³⁵

Although current stroke guidelines primarily emphasize absolute LDL-C targets, no clear consensus has been established on the clinical importance of relative reduction in LDL-C from baseline in patients with ischemic stroke. In a post hoc analysis of the TST trial, patients in the lower target group (<70 mg/dL) who achieved a $\geq 50\%$ reduction in LDL-C from baseline had a significantly reduced risk of major cardiovascular events compared with the higher target group (90–110 mg/dL) (HR 0.61, 95% CI 0.43–0.88, $p = 0.007$), whereas no significant benefit was observed among patients who failed to achieve a $\geq 50\%$ reduction despite reaching the lower LDL-C target (HR 0.96, 95% CI 0.73–1.26, $p = 0.75$).³⁵ Notably, although SPARCL did not define a specific LDL-C target, its reliance on high-intensity statin therapy inherently presupposed a substantial relative reduction in LDL-C, providing additional biological plausibility for the importance of the magnitude of LDL-C reduction.³⁴ In a post hoc time-varying analysis of SPARCL, patients achieving a $\geq 50\%$ reduction in LDL-C from baseline over time had a significantly lower risk of stroke (HR, 0.69; 95% CI, 0.55–0.87), whereas those with <50% LDL-C reduction did not demonstrate a significant risk reduction (HR 0.89, 95% CI 0.73–1.08), compared with patients with no change or an increase in LDL-C levels.³⁷ Consistent with these prior observations and extending the evidence beyond selected trial populations,^{7,34,35} our study demonstrated that achieving a $\geq 50\%$ reduction in LDL-C from baseline was independently associated with lower cardiovascular risk beyond merely reaching the target of <70 mg/dL. We further demonstrated that this association remained consistent when a more stringent LDL-C goal of <55 mg/dL was applied. Therefore, although not yet explicitly stated in the current stroke guidelines, our findings suggest that achieving a $\geq 50\%$ reduction in LDL-C from baseline may be an important consideration in lipid management strategies for secondary prevention in patients with ischemic stroke.

A critical debate in the management of patients with ischemic stroke centers on whether intensive LDL-C-lowering therapy is effective across all ischemic stroke subtypes or primarily in atherosclerotic stroke.³⁸ Although the SPARCL trial documented an overall benefit of atorvastatin 80 mg for patients experiencing recurrent stroke (HR 0.84, 95% CI 0.71–0.99) regardless of stroke etiology, the trial excluded patients with a cardioembolic source such as atrial fibrillation, and the risk

reduction was more pronounced in patients with carotid stenosis (HR 0.67, 95% CI 0.47–0.94) compared with those with no evidence of carotid stenosis (HR 0.90, 95% CI 0.74–1.08).³⁹ Similarly, the TST trial, which provided evidence for a favorable outcome with the lower LDL-C target, only enrolled ischemic stroke patients with evidence of atherosclerosis.³⁵ A meta-analysis from 11 randomized trials evaluating recurrent stroke risk found that the benefit of more intensive LDL-C lowering was only relevant for patients with ischemic stroke with evidence of atherosclerosis (RR 0.79, 95% CI 0.69–0.91) and the risk reduction was not significant in those without evidence of atherosclerosis (RR 0.95, 95% CI 0.85–1.07).³⁶ Owing to the inherent limitations of a study based on a health claims database, specific information on stroke etiology was unavailable in this study. Consequently, we were unable to assess the effect of $\geq 50\%$ LDL-C lowering according to stroke subtypes. In an attempt to partially mitigate this limitation, we performed subgroup analyses stratified by presumed cardioembolic status and formally tested for interaction. The association between achieving a $\geq 50\%$ LDL-C reduction and lower risk of the primary outcome remained directionally consistent regardless of presumed cardioembolic status, with no significant interaction detected (p for interaction >0.1), although point estimates were modestly attenuated in the cardioembolic subgroup. Further studies with adjudicated stroke subtype information are warranted to determine whether LDL-C-lowering strategies should be differentiated based on specific stroke etiologies, such as extracranial/intracranial stenosis or small vessel disease.³⁸ Ongoing randomized trials evaluating the benefit of LDL-C-lowering treatment in subgroups of ischemic stroke patients without atherosclerotic etiology may provide further insight and help inform precision medicine strategies for secondary stroke prevention.^{40,41}

This study has several strengths. First, we analyzed a large-scale nationwide cohort of patients with ischemic stroke over a mean follow-up of 5.72 years, incorporating a comprehensive range of variables including lipid parameters, traditional vascular risk factors, socioeconomic status, lifestyle factors, and disability status. Second, LDL-C levels were modeled as time-varying covariates, which allowed the analysis to capture dynamic patterns of LDL-C control in a real-world clinical setting. Despite these strengths, several limitations of this study should also be acknowledged. First, because the health claims database was designed primarily for reimbursement purposes, it lacks detailed clinical information such as infarct size, location of the lesion, severity of the stroke, presenting symptoms, and specific etiology of the stroke, all of which can act as potential confounders. Second, although participation in the national health examination program is relatively high, a selection bias may exist because our analysis was limited to patients who underwent health examinations. Moreover, even among those included, long intervals between examinations limited the accurate assessment of longitudinal LDL-C trajectories. Indeed, the average number of LDL-C measurements per patient was low (1.53), limiting our ability to track

continuous longitudinal changes. Furthermore, the time to LDL-C target attainment after the index stroke could not be reliably assessed, as the mean interval from the index stroke to the first poststroke LDL-C measurement was relatively long; this represents an inherent limitation of using biennial health examination data to evaluate the rapidity of LDL-C lowering in clinical practice. Third, we did not account for the specific type of statin or other lipid-lowering agents used for treatment. Nonetheless, our analysis focused on the magnitude of the reduction in LDL-C itself, which previous studies have suggested is a key determinant of the reduction in cardiovascular risk, irrespective of the specific lipid-lowering agents used.^{36,42} Fourth, achieving appropriate levels of LDL-C may reflect a greater adherence to comprehensive secondary prevention strategies, including blood pressure control, antithrombotic therapy, and lifestyle modification. By contrast, a substantial reduction in LDL-C can also reflect poor nutritional status or underlying health deterioration in some patients. Fifth, although operational definitions of stroke and MI based on Korean health claims data have been validated, the possibility of misclassification of outcomes cannot be entirely excluded. Nevertheless, we observed a consistent and robust association between a $\geq 50\%$ reduction in LDL-C and all-cause mortality, an outcome verified through Statistics Korea, which supports the overall reliability of our findings. Sixth, as this is an observational cohort study, causal inference cannot be drawn from these findings. Residual confounding remains possible despite multivariable adjustments, and our results should be interpreted as hypothesis-generating rather than providing definitive evidence for the revision of clinical guidelines. Finally, this study was conducted exclusively in Korea and a Korean statistical report indicated that large artery atherosclerosis was the most common subtype (36.1%) in this population.⁴³ Therefore, further validation is necessary to determine the generalizability of our findings to other ethnic groups and Western populations.

In conclusion, achieving a $\geq 50\%$ reduction in LDL-C from baseline was independently associated with a lower risk of secondary cardiovascular events in patients with ischemic stroke and offers advantages beyond simply reaching an absolute LDL-C goal. Notably, our real-world data revealed that a large proportion of ischemic stroke patients still exhibit suboptimal LDL-C control. In clinical practice, strict monitoring and intensive LDL-C control strategies may be helpful for minimizing the residual risk of recurrence and improving patient prognosis.

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Author Contributions

M. Baik: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. J. Jeon: drafting/revision of the manuscript for content, including medical writing for content; analysis or

interpretation of data. J. Yoo: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. J. Kim: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data.

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